

UI Wireframes and Screen Behavior v2.0

1. Main Screen

| Alcohol Label Verification App |

| Batch-first OCR-assisted label review prototype |

Step 1: Upload Application Records

[Upload CSV File]

Required CSV fields:

record_id, brand_name, class_type, alcohol_content, net_contents, government_warning

Optional CSV fields:

bottler_producer, country_of_origin, beverage_type

Step 2: Upload Label Images

[Upload PNG, JPG, or JPEG files]

Supported files:

PNG, JPG, JPEG

Step 3: Verify Labels

[Run Verification]

Batch Progress

Total Labels: 0

Processed: 0

Needs Review: 0

Failed: 0

Passed: 0

Results Dashboard

| Image Name | Matched Record | Match Confidence | Overall Status | Processing Time |

|-----|-----|-----|-----|-----|

[Export Results CSV]

2. Required UI Principles

The interface must be:

- Simple
- Obvious
- Low clutter
- Large enough to read
- Friendly to nontechnical users
- Clear about what action to take next
- Batch-first
- Minimal manual data entry
- Transparent about uncertainty

The user should never need to hunt for the next button.

3. Main User Flow

1. User uploads an application CSV.

2. Application validates required CSV columns.
 3. User uploads one or more label images.
 4. User clicks Run Verification.
 5. Application processes labels one at a time.
 6. Results appear as each label completes.
 7. User reviews match confidence and validation results.
 8. User may review extracted OCR text if needed.
 9. User exports results as CSV.
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4. CSV Upload Behavior

Valid CSV

If the uploaded CSV includes all required fields, the application displays:

CSV uploaded successfully.

Records loaded: [number]

Required fields:

- record_id
- brand_name
- class_type
- alcohol_content
- net_contents
- government_warning

Optional fields:

- bottler_producer
 - country_of_origin
 - beverage_type
-

Missing Required CSV Fields

If required fields are missing, processing stops.

Example message:

The uploaded CSV is missing required fields: brand_name, alcohol_content.

Please upload a CSV with the required template format.

Extra CSV Fields

Extra fields should not break the application.

The application should safely ignore unrecognized columns.

5. Image Upload Behavior

Users may upload one or more label images.

Supported types:

- PNG
- JPG
- JPEG

If an unsupported file type is uploaded, show:

This file type is not supported. Please upload a PNG, JPG, or JPEG image.

6. Batch Verification Screen

Batch Verification

Application Records Loaded: 12

Label Images Uploaded: 8

[Run Verification]

Processing Status

Currently Processing:

IMG_4829.png

Completed:

3 of 8

Total Batch Time:

9.4 seconds

Results

Image Name	Matched Record	Match Confidence	Status	Time
-----	-----	-----	-----	-----
IMG_4829.png	Wittner Brewery IPA	97%	Pass	2.3s
IMG_4830.png	Thomas Hill Winery	81%	Needs Review	3.0s
IMG_4831.png	No Match	42%	Needs Review	2.7s

[Export Results CSV]

7. Result Status Display

Pass

Show when detected information matches expected application data.

Meaning:

The system found no mismatch in the validated core fields.

Fail

Show when detected information clearly conflicts with expected application data.

Example:

Expected ABV:

45%

Detected ABV:

40%

Missing

Show when a required field was not found.

Example:

Government warning not detected.

Needs Review

Show when the application cannot confidently decide.

Examples:

- OCR uncertainty
- Ambiguous record match
- Poor image quality
- Low confidence score
- No clear matching application record

Needs Review should not be treated as a failure.

It means a human reviewer should inspect the label.

8. Record Matching Display

Each processed label should show:

- Image name
- Suggested matched record
- Match confidence
- Match status
- Notes

Match Statuses

Auto Match

Confidence:

95% and above

Meaning:

The application confidently matched the image to a CSV record.

Needs Review

Confidence:

70%–94%

Meaning:

The application found a likely match but recommends human confirmation.

No Match

Confidence:

Below 70%

Meaning:

The application could not confidently match the image to a CSV record.

9. Field-Level Results View

Each label result should allow the user to view field-level results.

Image:

IMG_4829.png

Matched Record:

Wittner Brewery IPA

Overall Status:

Pass

Processing Time:

2.3 seconds

Field Results

Field	Expected	Detected	Status	Method
-----	-----	-----	-----	-----
Brand Name	Wittner Brewery IPA	WITTNER BREWERY IPA	Pass	Fuzzy
Class / Type	India Pale Ale	India Pale Ale	Pass	Fuzzy
Alcohol Content	9%	9%	Pass	Exact
Net Contents	12 fl oz	12 fl oz	Pass	Exact
Government Warning	Required warning text	Required warning text	Pass	Strict

[View Extracted OCR Text]

10. Optional OCR Text Panel

Include a collapsible section:

View Extracted OCR Text

Purpose:

Allows technical reviewers and compliance users to see what the system read from the image.

This supports transparency and troubleshooting.

11. Single Label Test Mode

Single Label Test Mode is secondary.

Purpose:

- Demonstrations
- Troubleshooting
- Isolated testing

Screen:

Single Label Test Mode

[Upload Label Image]

[Enter Expected Values Manually]

Brand Name

[_____]

Class / Type

[_____]

Alcohol Content

[_____]

Net Contents

[_____]

Government Warning

[_____]

[Verify Label]

This mode should not be presented as the main workflow.

12. Error Messages

Error messages should be plain English.

Examples:

No CSV file was uploaded. Please upload an application records CSV first.

No image was uploaded. Please choose at least one label image.

The image was uploaded, but no readable text was detected. This label has been marked Needs Review.

The uploaded CSV is missing required fields: record_id, brand_name.

This file type is not supported. Please upload a PNG, JPG, or JPEG image.

13. Processing Behavior

Labels are processed sequentially.

Results should appear as each label completes.

This avoids leaving the user staring at a blank screen during batch processing.

Sequential processing is intentionally chosen for the prototype because it is:

- Simple
- Transparent
- Easy to debug
- Responsive for small batches

Future production versions may use:

- Background workers
- Queue-based processing
- Parallel processing
- Larger batch support

14. Export Results

The user can export verification results as CSV.

Export may include:

- Image Name
- Matched Record ID
- Matched Brand Name
- Match Confidence
- Overall Status
- Field Name
- Expected Value
- Detected Value
- Field Status
- Processing Time
- Notes

Purpose:

- Supervisor review
- Compliance documentation
- Follow-up review
- Audit preparation

15. Accessibility and Usability Notes

The interface should prioritize:

- Clear section labels
- Large buttons
- Short instructions
- Minimal scrolling where possible
- Plain-language status messages
- Avoidance of technical jargon
- Clear color/status indicators
- No hidden critical actions

The application should be understandable to both technically comfortable users and users with limited software experience.